FEEDBACK:

[redacted] this was an excellent essay; you've clearly researched this topic widely and have drawn on a really good range of examples within this word count and they were nicely introduced. Some minor comments to help with revision alongside ways to approach essays in CBN are inscribed in the margins. Some good critical engagement throughout this essay. The main feedback I am finding myself giving across this topic is to do with how to approach answering the exact question the essay title asks. Then, each time you present a piece of evidence that has advanced our understanding of whatever the Essay Q is asking make sure to outline what, exactly, it demonstrates.

For example, rather than stating that a study 'highlights differences between different types of memory' explain which functions are dissociable / what evidence supports that distinction / which brain regions are implicated / what limitations remain in interpreting the findings etc. If we take the example of Korsakoff syndrome vs other amnesic syndromes (like H.M's case) – this demarcates the dependence of episodic memory on hippocampal-diencephalic circuitry (not just hippocampus in isolation). Patients with this present with profound amnesia but also exhibit confabulation – and these things often sit alongside relatively preserved immediate memory span. We know that the neurodegeneration in this instance (from deficiency of thiamine) predominantly involves the mammillary bodies and medial thalamus; the resulting impairment in episodic encoding and retrieval (even in the absence of direct or extensive hippocampal damage!) demonstrates that declarative memory depends on distributed rather than exclusively hippocampal circuitry. Occurrence of confabulation further implicates dysfunction of frontal-diencephalic networks involved in contextual and reality monitoring (which shows us that accurate episodic remembering also depends on mechanisms for monitoring the source, context and veracity of retrieved information). The differing cognitive phenotypes of amnestic syndromes -- including bilateral medial temporal lobe amnesia (as in H.M.), transient global amnesia, transient epileptic amnesia and amnestic Alzheimer's disease -- reflect the involvement of different components of memory systems in encoding/consolidation/retrieval (& variation in the onset, duration and pattern of impairment also provides evidence for the relative functional specialisation of hippocampal & neocortical networks).

More generally, if we consider broader classes of memory disorders like semantic dementia [see Patterson et al.'s review (2007) – would recommend as revision], we can use it to question what's going on in patients who progressively lose knowledge about objects and concepts -- and how can we tell that they're not simply experiencing difficulties retrieving their names (which would just be classed as anomia; i.e. failure to retrieve the name of an object is distinct from degradation of the semantic representation of the object itself). In an example like that, you could discuss how this impairment extends across different tasks and sensory modalities (e.g. picture naming, object recognition and conceptual associations), despite other cognitive abilities remaining relatively preserved early in the disease.

What do memory disorders tell us about the organisation of human memory?

Introduction

Memory disorders provide critical information about the organisation of human memory. Behavioural observations and task performance paired with the identification of damaged structures gives insight to the anatomical organisation of memory, particularly in highlighting double dissociations between the various domains of memory. In this way, we can distinguish memory temporally and based on what cognitive activity it contributes to.

Short term memory and working memory

Short term memory (STM) refers to the limited span of information that can be held for twenty to thirty seconds. It is closely linked to working memory (WM), as information from the STM is stored and manipulated to perform a task, such as retaining the details of instructions. Within this type of memory, there are different modalities, such as verbal, spatial and auditory, and different brain regions may be responsible for supporting them. This is supported with observations from clinical reports of distinct auditory and visual working memory deficits being associated with brain lesions with different locations (Butters *et al.*, 1970). Additionally, a neuroimaging study where participants had to perform identity and location tasks with a delay supports the involvement of distinct areas in the prefrontal cortex for each type of task, though some areas were involved in both (Courtney *et al.*, 1996). A model that is supported by these findings is the Baddeley and Hitch multicomponent model of WM. They describe a phonological loop for auditory information and a visuospatial sketchpad for visual information that are bound together by an episodic buffer, all regulated by a supervisory system.

Commented [EC1]: When approaching essays, I generally recommend first defining the terms the essay Q is asking about (here: (1) *memory disorders* and (2) *organisation of memory*) to structure your answer. I think you do this quite well, but you could have looked more closely at distinct memory disorders themselves and contrasted them in terms of deficits, brain regions affected & ways we've tested this. This was a deliberately broad essay Q but other exam Qs on this topic could ask specifically about cases where we see amnesia or confabulation -- or could ask about disorders of episodic memory -- or associative agnosia or semantic dementia specifically. So be sure to make note of the specific ask in the title.

Commented [EC2]: yes -- that temporary retention of a limited amount of information that we consider to be 'short term memory' is typically only a few seconds. This often-cited 20-30s comes from those classical experimental findings by Peterson (1959) that showed rapid forgetting when rehearsal was prevented. But ! We now know that retention depends on a number of other things: rehearsal, interference, attention and task demands. So, there's not a universal 30-second cut-off after which we consider something to not be 'short-term' memory.

Commented [EC3]: Note that the phonological loop supports verbal/phonological information (not exclusively auditory information). The 'episodic buffer' that you mention briefly here is something that Alan (Baddeley) proposed, a bit later (Baddeley, 2000) - which was this idea that there needs to be something that INTEGRATES information across modalities and with LTM for this to be useful.

Commented [EC3R2]: So let's say your friend witnesses a crime and then tells you:

"I saw him by the tree over the street. He was wearing a dark green jacket and had in his left hand a knife and, in his right, a Sainsbury's Bag for Life." For you, understanding and temporarily remembering that involves integrating verbal information, visual imagery, spatial relationships and existing semantic knowledge into a *unified representation* of an event. The 'episodic buffer' is the proposed workspace in which this integrated representation can be temporarily maintained and held for a while.

Long term memory

While STM is limited in its capacity and the time it can be retained, long term memory (LTM) is a much longer lasting and expansive form of information storage. The case of patient H.M. points to a double dissociation between these types of memory (Eichenbaum, 2011), which posits that they each have their respective mechanisms and anatomical regions in the brain. H.M. underwent bilateral medial temporal lobe (MTL) resection to treat his severe epilepsy, and one of the main changes observed in his behaviour afterwards was his severe anterograde amnesia. Despite his good overall cognition, perception and vision, he unable form any new long-term memories. Strikingly, he could perform brief tasks, indicating an intact STM and WM, but once he was distracted, he could not remember being given any instruction. Overall, his case suggests that STM and LTM are anatomically distinct, with MTL structures such as the hippocampus being crucial in supporting LTM.

Commented [EC4]: Not quite. H.M. alone demonstrates a single dissociation: relatively preserved immediate memory despite severely impaired formation of new declarative long-term memories. A double dissociation requires complementary evidence from patients such as K.F., with disproportionately impaired verbal STM and relatively preserved LTM

Commented [EC5]: when writing about amnesia consider what the different types we observe tell us about the brain regions and processes involved.

The distinction between anterograde and retrograde amnesia gives us really useful evidence for the anatomical and temporal organisation of human memory! The profound anterograde amnesia that H.M. presented with shows us that the medial temporal circuitry arround the hippocapus (particularly the entorhinal, perirhinal and parahippocampal cortices) has this totally crucial role in encoding and binding the spatial, temporal and contextual features of new experiences into retrievable episodic representations.

Commented [EC5R2]: Retrograde amnesia, by contrast, we define as the loss of memories acquired BEFORE neurological injury -- you hear often about the temporal gradient of this amnesia (i.e relatively recent memories are disproportionately impaired while more remote memories remain accessible). This patterning, consistent with Ribot's law, has led to systems consolidation theory, which argues that newly acquired memories initially depend heavily on hippocampal-neocortical interactions but *gradually* become supported by more distributed neocortical representations :)

Commented [EC6]: Ah -- well! Not entirely, one of the things to discuss.

Commented [EC6R2]: H.M. retained the ability to acquire certain procedural skills and showed other forms of non-declarative learning. The impairment was predominantly in forming new *declarative memories*.

Papez circuit

Aside from the hippocampus, other structures that form a circuit are also implicated in memory. Some of the components of the Papez circuit, shown in a schematic in Figure 1, are involved in the memory system. It follows that damage to any of them impairs memory. Patient N.A. demonstrates this, as he suffered severe anterograde and retrograde amnesia following an injury to his left dorsomedial thalamus. Additionally, Korsakoff's patients, whose untreated thiamine deficiency primarily damages the dorsomedial thalamus and mamillary bodies, also suffer severe retrograde and anterograde amnesia. Therefore, it is not the hippocampus alone that supports human memory.

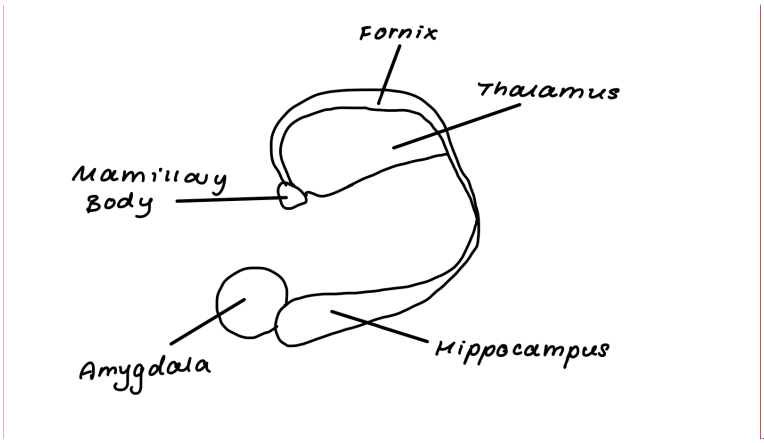


Figure 1: The memory related components of the Papez circuit. Based on diagram in (Bear, 2026).

Consolidation and recall

Within LTM, the consolidation and the recollection of memories are separate abilities. H.M. appeared able to recall some of his remote childhood memories, although further examinations indicated that he had retrograde amnesia that spanned memories over decades prior. One explanation of this is the consolidation theory, which states that remote declarative memories are initially stored in the hippocampus and cortical regions before being transferred to the neocortex for permanent storage. This fits the observation that H.M. could recall some of his most remote memories as his neocortex would have been spared during his procedure. An alternate idea is the multiple trace theory, where experiences form traces whenever they are re-encountered. Older experiences have more traces, as they have had more time to be

Commented [EC7]: Often, yes, but lesion location, extent, laterality and connectivity all matter here. Not every clinical event that affects a structure associated with the Papez circuit (be it a tumour etc.) necessarily produces clinically significant amnesia.

Commented [EC8]: yes, N.A.'s injury involved the left diencephalon, with evidence implicating the dorsomedial thalamic region and adjacent structures/white matter

Commented [EC9]: important to distinguish thiamine deficiency, Wernicke encephalopathy, etc. Will discuss in tutorial.

Commented [EC9R2]: Confabulation -- which is v classic in alcohol related dementais like Korsakoff patients -- is important to distinguish from normal amnesia. When we talk about confabulation, we're referring specifically to production of false or distorted memories without deliberate intent to deceive; patients may (really quite confidently) describe events that never occurred or misattribute genuine experiences to the wrong time or context. In Korsakoff's patients, that thiamine deficiency damages structures within the diencephalic memory network as you describe here (mammillary bodies and medial thalamic nuclei), disrupting their connections with the hippocampus and frontal cortex. So THAT -- how that causes the memory phenotype we observe -- is key.

Aka: this contributes to impaired episodic memory, *and at the same time*, dysfunction of frontal monitoring and strategic retrieval processes can *prevent patients from recognising that a recollection is inaccurate* or inappropriate to the present context.

Commented [EC10]: ah, but do consider that amnesia and confabulation are not equivalent: patients can have severe amnesia without confabulating, indicating that failures of memory retrieval and failures of memory monitoring are at least partly dissociable

Commented [EC11]: excellent

revisited. Regardless of how memories are consolidated, there also seems to be a double dissociation with how they are recalled as findings from Korsakoff's patients show no correlation in the severity of their anterograde and retrograde amnesia. Consequently, the processes governing long-term memory formation and recall respectively must be distinct as they are not impacted equally by the same damage.

Explicit and implicit memories

Another important distinction is that between explicit and implicit memories, also known as declarative and non-declarative. While explicit memories require the conscious recollection of a fact, implicit memories are encoded without a conscious awareness of the learning process. Within explicit memory, episodic memory refers to the recollection of autobiographical events that are related to an awareness of temporal and spatial context, whereas semantic memory refers to retrieval of factual information without the need for context. Amnesic patients commonly suffer with the formation and recollection of both types of explicit memory, ~~but the case of patient K.C. supports a double dissociation between them~~. After complete destruction of his left hippocampus, parahippocampal gyrus, entorhinal and perirhinal cortex, he lost most of his autobiographical episodic memories but had retained more semantic memories that he had learned in his education and work prior. Tying in with the multiple trace theory, it could be that semantic facts are simply encountered more frequently than episodic facts. For example, patient H.M. had been able to correctly identify the surnames of some figures who had become famous after his operation with their first names as cues. However, this could be an example of priming, which has been showed to be spared in amnesiacs, including H.M., with word stem completion tests and dot tasks.

Procedural memory, a type of implicit memory, is supported by perceptuomotor skills and is obtained slowly with gradual learning after repetition and practice. There are strong indications that it is not governed by the same areas responsible for explicit memory. The contrast between the performance of Huntington's and Korsakoff's patients in tasks such as mirror-reading and pursuit rotor demonstrates this. Korsakoff's patients show a gradual improvement, despite their diencephalic damage, while in Huntington's patients, whose basal ganglia are mainly affected, performance remains stable. Furthermore, amnesic patients can also improve their performance in mirror-reading, while remaining poor at recognising familiar words from the task (Cohen and Squire, 1980). Together, these solidify a double

Commented [EC12]: Well, you could argue this but a lack of correlation is not a double dissociation and does not establish that two processes are independent. Differences in impairment could reflect: lesion heterogeneity; measurement reliability; premorbid factors; different task demands..

Commented [EC13]: Explicit memory involves conscious/declarative access; implicit memory is demonstrated through changes in performance without requiring conscious recollection

Commented [EC14]: So not quite. K.C. provides evidence for disproportionately impaired episodic (autobiographical) memory relative to semantic knowledge. This is a single dissociation. A complementary pattern, such as impaired semantic knowledge with relatively preserved episodic memory in some semantic dementia patients, is needed to argue for a double dissociation.

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Commented [EC15]: For me to check. As I remember K.C. sustained widespread brain injury following a motorcycle accident, including BILATERAL medial temporal and other damage... His memory phenotype can't be attributed to complete destruction of these structures specifically on the left.

Commented [EC16]: To finish my thread earlier -- cases of confabulation clinically - whether it be in hepatic encephalopathy, Korsakoffs, Alzhiemer's - are useful in that they suggest that episodic remembering involves interacting systems responsible for RECONSTRUCTING past experiences, establishing their temporal and contextual relevance, and (crucially!
evaluating whether retrieved information is accurate. So confabulation is a nice tidy illustration of how memory disorders can reveal the organisation of the processes underlying remembering, rather than merely identifying where memories are stored (brain structure wise).

Commented [EC17]: Neurodegenerative diseases reveal how cognitive phenotypes reflect the anatomical distribution of their underlying neuropathology and related neurodegeneration. Consider the patterning of where this pathology first accrues, its corresponding impairment, and how the temporal progression of neurodegeneration maps to clinical deficits seen in the disease. Braak & Braak in 1991 showed that Alzheimer's disease (where the aggregation of amyloid-β and tau causes localised neurodegeneration) compromises hippocampal-entorhinal networks early in the disease (impairing new episodic learning and retention) before extending into broader association cortices as the disease progresses.

You can contrast that to Huntington's disease (HD), where we know that mutant huntingtin forms soluble oligomers and insoluble aggregates, including neuronal intranuclear inclusions- the neurodegeneration that results from that causes disruption of frontostriatal circuitry & impairs strategic encoding and retrieval (patients display impaired spontaneous recall but relatively better performance with recognition or semantic cueing!) before widespread cortical degeneration develops.

dissociation between explicit memory processes in the MTL and implicit memory processes elsewhere, such as the frontal cortex, basal ganglia, parietal and cerebellar regions.

Challenges

Learning about memory through its disorders is not flawless. A major concern is the overlap between memory and other cognitive processes, such as language. These can be difficult to disentangle clinically, for example a study on the subjective experience of aphasic patients identified that some participants used the language of memory to describe their difficulties in word-finding, although they were truly referring to a failure of lexical access or retrieval (Fama, Lemonds and Levinson, 2022). Another concern is that changes in memory performance are associated with mood. Self-reported measures of anxiety have been reliably related to poorer performance in tasks testing working memory (Moran, 2016), and depression has been associated with a small to moderate deficit in working memory (James *et al.*, 2021). It may be challenging to account for mood changes between participants studied for their memory disorders, in part because mood may fluctuate or also be affected by their conditions, such as by frustration of memory loss. Moreover, the damage caused by lesions or conditions across individuals is not uniform as there may be slight variations in the sizes and areas that are affected. All of these could lead to observable behaviour changes even between people with a seemingly similar condition.

Conclusion

In conclusion, observations from lesions and conditions that affect memory give vital clues about its organisation in humans. Evidently, the structures that support various aspects of memory are widespread in the brain, with different areas connecting to each other and supporting different faculties, as none of the disorders mentioned lead to a loss of all types of memory. Neuropsychological studies of memory disorders still have their limitations with regards to the study of memory but are important in conjunction with other findings from studies with healthy participants and neuroimaging studies.

Word count = 1361

Bibliography

Commented [EC18]: Nice.

Commented [EC19]: excellent example

Commented [EC20]: I really liked your critical engagement in this paragraph of what confounding factors might influence how we study memory

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